

Dimension Deck GDD



Tecnológico
de Monterrey

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Dimension Deck — Game Design Document

Course: Software Construction and Decision Making

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1. Game Design

1.1 Summary

Dimension Deck is a retro-styled top-down action roguelite where every room of the map hides a surprise: a new card to collect, a hidden reward, or an unexpected boss fight. The player builds their deck as they explore two distinct dimensions — the Dark Ages and the Old West — finding attack, defense, and support cards scattered across the world, creating a unique and evolving strategy with every run. Combat is entirely card-based, meaning the hand the player has built determines whether they survive what lies behind the next door.

With a world that shifts themes and challenges as the player progresses and a randomized structure that ensures no two runs ever feel the same, Dimension Deck delivers the tension of a roguelite wrapped in the charm of classic pixel-art adventure.

Dimension Deck sits at the intersection of two genres without belonging fully to either. Unlike *Slay the Spire*, where combat is turn-based and purely strategic, and unlike *Neon White*, where cards are disposable movement tools, Dimension Deck makes cards and real-time movement equally necessary at all times. The player cannot win by standing still and thinking, and cannot win by running without a plan. The closest comparison is: *The Binding of Isaac* meets *Slay the Spire*, but where the deck actively fights back through automated synergies without the player pressing a button.

1.2 Gameplay

Dimension Deck is a dynamic game that combines a fast-paced action roguelite with the freedom of a deck builder using a 2D top-down design. It borrows movement and perspective from *The Binding of Isaac* while incorporating the strategic deck management of *Slay the Spire*.

Players use their deck of cards to materialize attacks, defenses, and effects in real time. The combat system uses two categories of cards: **Active Cards**, which the player triggers manually to attack, heal, or defend; and **Automatic Cards**, which

activate when certain in-game conditions are met, functioning as passive synergies.

Players navigate randomized rooms to find loot, shops, or enemies, advancing toward the goal of clearing each room and defeating the boss of every dimension to progress.

Generation Rulebook

Each dimension is built from a sequence of rooms generated at the start of every run. A dimension always contains a minimum of 8 and a maximum of 12 rooms before a Boss node is placed at the end. The exact count is chosen randomly within that range each time, so the player never knows how close they are to the boss until they reach it.

Rooms are not placed in a straight line. The generator builds a small branching map where the player may have two or three possible paths to take at certain points, but all paths eventually converge at the Boss node.

Room Type Weighting

Room Type	Base Chance	Notes
Combat Room	60%	Always available
Chest Room	15%	Drops one card on clear
Shop Room	10%	Increases over time (see below)
Shrine Room	10%	Offers one card upgrade trade
Glitch Room	5%	Contains a Glitch Pillar

Shop Room Scaling: The chance of a Shop Room appearing increases by 10% for every 3 consecutive combat rooms cleared without visiting a shop. This bonus stacks and resets to 0% the moment a Shop Room is entered, preventing the player from going deep into a dimension without access to cards or upgrades.

1.3 Mindset

The goal is for the player to feel overwhelmed by the number of enemies on screen, but empowered by the automated synergies their deck produces. The tension comes from managing movement and card rotation at the same time. A

good run should feel like controlled chaos, where the player is always one smart card play away from turning a losing fight around.

The emotional target is: *"I was about to die, but my deck saved me."*

2. Technical

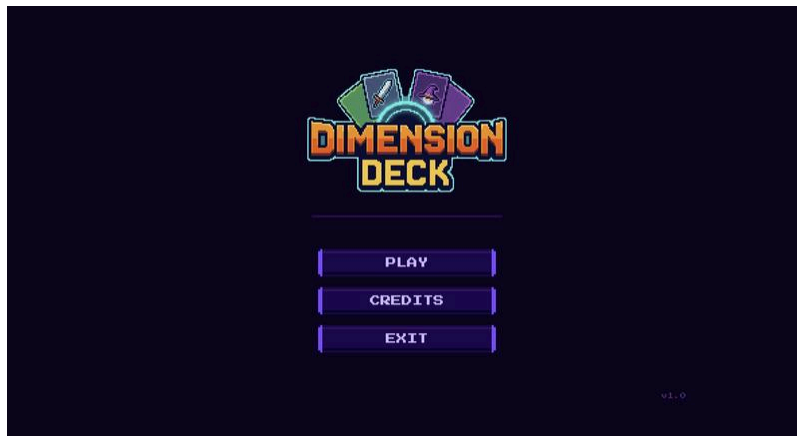
2.1 Screens

Dimension Deck uses several core screens organized into three categories: **Navigation & Progression**, **Gameplay & Customization**, and **Economy & Results**.

Navigation and Progression Screens

Start Game Screen

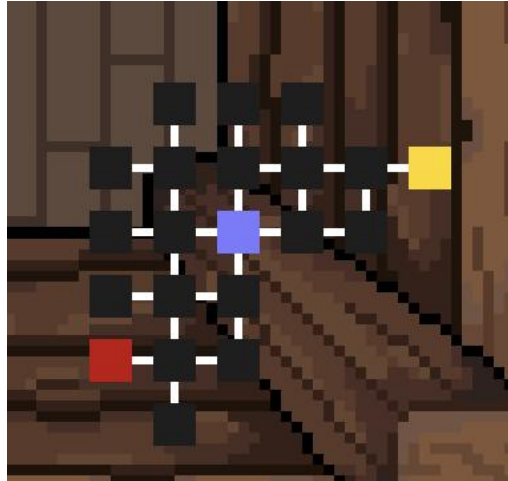
- Purpose: Entry point to the game; the player selects their character and starts a run.
- Elements: Character selector (Knight or Cowboy), Start button, and back navigation.



Start Game

Dimension Map

- Purpose: A full view of the current dimension, allowing the player to choose their path through rooms.
- Elements:
 - Nodes representing room types (Combat, Chest, Shop, Shrine, Boss).
 - Lines showing valid movement between nodes.
 - Mini-map toggle via the **M** key.

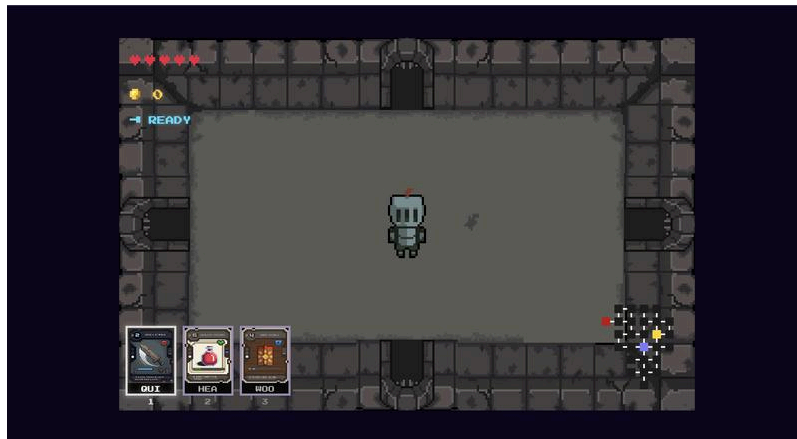


Dimension Map

Gameplay and Customization Screens

Room Screen (Gameplay)

- Purpose: The screen where combat takes place in real time.
- Elements:
 - Player and enemy health bars.
 - Active card hand displayed at the bottom center.
 - Shield and credit counters.



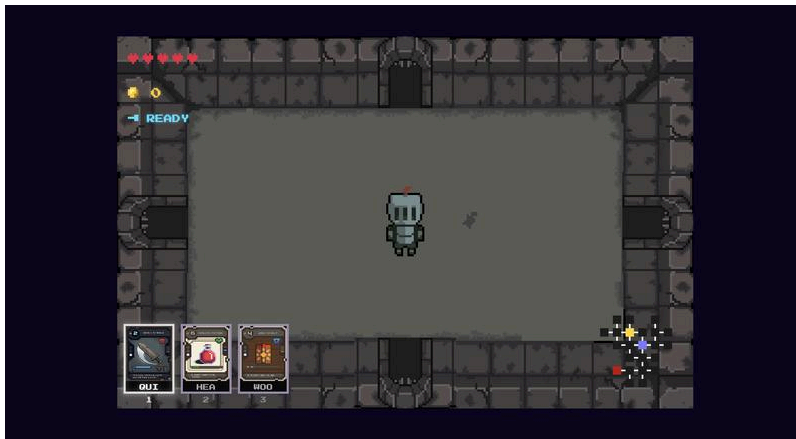
Gameplay HUD

Combat in Action



Combat

Dark Ages Dimension



Dark Ages

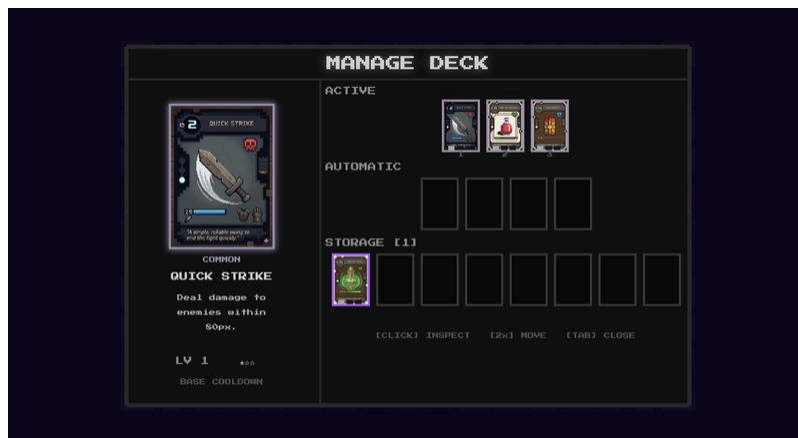
Old West Dimension



Old West

Deck Management Screen

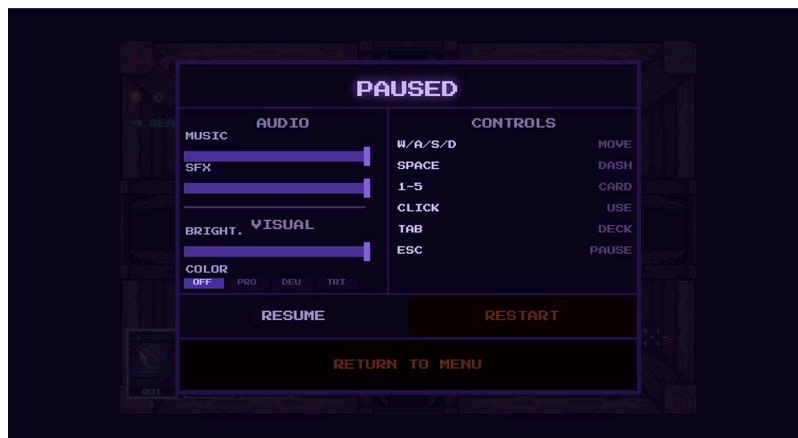
- Purpose: Allows the player to review and manage their full deck at any time.
- Elements:
 - Grid view of all cards currently in the player's deck.
 - Filter and sort options by rarity and type.
 - Slot counts for active and automatic card pools.



Manage Deck

Pause Menu

- Purpose: Pauses the run and provides options to resume or quit.

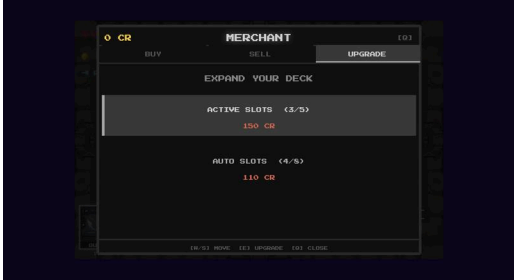


Paused

Economy and Result Screens

Shop Screens

- Purpose: The player spends earned dimensional credits to buy cards, sell cards, or upgrade card slots.

Buy	Sell	Upgrade Slots
		

2.2 Controls

Movement and Navigation

Key	Action
W / Arrow Up	Move up
A / Arrow Left	Move left
S / Arrow Down	Move down
D / Arrow Right	Move right
Mouse Cursor	Aim at enemies or target card destinations
M (tap)	Toggle mini-map overlay
M (hold)	Open full-screen dimension map

Card Actions

Key	Action
1	Execute first active card (leftmost slot)
2	Execute second active card
3	Execute third active card
4	Execute fourth active card (requires slot expansion)
5	Execute fifth active card (requires slot expansion)
Left Click	Confirm and execute the selected card at cursor position

Combat and Utility

Key	Action
Spacebar	Dodge or dash in the current movement direction
E	Interact with chests, Glitch Pillars, and the shop
Tab	Toggle deck management overlay
Escape	Pause the game or close any open menu

Technical UX Notes

- **Smart Mapping:** When a card is played, keys shift left to eliminate dead slots in the middle of the hand.
- **Map Persistence:** Room types are permanently revealed once the player enters each room, even on backtrack.
- **UI Tooltips:** In the deck management screen, hovering over a card displays its full stats.

2.3 Cards

2.3.1 Card System

Cards are the foundation of combat and progression in Dimension Deck. Every offensive and defensive action is executed through the player's deck, which is divided into two categories.

Active Cards (Manual Execution)

Active cards are weapons the player triggers manually to deal damage, heal, or generate shields.

- **Capacity:** Starts at 3 slots, expandable to a maximum of 5.
- **Rotation:** Once played, the card enters a cooldown period before returning to the hand. This prevents spamming a single card and forces thoughtful timing.

Automatic Cards (Trigger-Based)

Automatic cards sit in a secondary pool and activate when specific combat conditions are met, functioning as passive synergies.

- **Capacity:** Supports between 4 and 8 slots.

- **Function:** These cards reward synergistic deck-building without requiring player input.

Card Rarities

Rarity	Color	Description
Common	Gray	Base-level cards found frequently. Starting deck is composed of these.
Uncommon	Green	Slightly enhanced cards with improved values.
Rare	Blue	Improved cards with higher damage, healing, or utility.
Epic	Purple	Powerful mechanics that serve as the cornerstone of a specific build strategy.
Legendary	Gold	Game-changing cards that fundamentally alter how the player approaches every run.

2.3.2 Card Resistance System

Expanding the player’s deck increases their power, but also activates the Card Resistance system. As more card slots are unlocked, enemies gain resistance to card damage, forcing the player to keep evolving their strategy throughout the run.

Resistance Scaling Formula:

|

$$Resistance \% = (Active\ Card\ Slots - 3) \times 5\%$$

Active Card Slots	Resistance Applied
3 (base)	0%
4	5%
5 (max)	10%

The automatic card pool adds its own separate resistance layer:

|

For every 2 automatic slots above the base 4, enemies gain an additional 3%.

Automatic Card Slots	Additional Resistance
4 (base)	+0%
6	+3%
8 (max)	+6%

Both values stack. A player with 5 active slots and 8 automatic slots faces a total of **16% damage reduction** on all enemies.

2.3.3 Card Acquisition

Field Rewards

- **Combat:** Defeating the final boss of each dimension grants a choice of 3 cards from a randomized selection.
- **Treasure Chests:** Hidden throughout procedural floors; contain a randomized card.

The Dimension Store (Shop)

- Earned dimensional credits can be spent on a rotating stock of 5 cards per visit.
- Players can also sell unwanted cards and purchase slot upgrades.
- Prices range from 65 to 380 credits depending on rarity.

Glitch Pillars

Hidden objects scattered across rooms that do not appear on every run. When the player touches a Glitch Pillar, it rewards them with a single card pulled randomly from any rarity tier. The card goes directly into the deck — no selection screen.

2.3.4 Card Upgrades

Every card has three levels: **Base**, **Upgraded**, and **Max**.

- Acquiring a second copy of the same card merges them automatically, advancing to Upgraded.
- Acquiring a third copy advances to Max.
- Additional copies of a Max-level card convert into dimensional credits based on rarity:

Rarity	Credits Returned
Common	10
Rare	25
Epic	50
Legendary	100

Active Cards - *Upgraded*: Cooldown reduced by 30%. - *Max*: Base effect value increased by 40%; unlocks a secondary effect.

Automatic Cards - *Upgraded*: Trigger condition widens, activating in more situations. - *Max*: Potency of the effect increases (larger damage bonus, longer duration, or stronger status).

2.3.5 Combat: Card Rules

Active Card Execution

- **Instant Activation:** Pressing the corresponding number key triggers the card's effect immediately. There is no cast time or mana cost.
- **Cooldown State:** After activation, the slot dims and displays a recharge animation. The player cannot use that slot until the timer finishes.
- **Variable Recovery:** Common cards have short cooldowns (2–6 s); Legendary cards have long cooldowns (10–18 s).

Damage and Collision

- Active melee cards deal damage to all enemies whose hitbox falls within a cone-shaped arc in the aim direction.
- Drain cards find and damage the nearest living enemy.
- Defensive cards apply a flat shield or invincibility frames instantly with no directional requirement.

Hand Cycling

Since there is no mana limit, the primary skill gap is rotation management. Players must cycle through their 1–5 keys to ensure an offensive or defensive option is always available.

2.3.6 Type of Cards

Active Cards

#	Name	Rarity	Cooldown	Effect
1	Quick Strike	Common	2 s	Deal 25 damage to enemies within 80 px in a forward cone
2	Heal Pulse	Common	5 s	Restore 25 HP
3	Wood Shield	Common	6 s	Absorb the next 20 damage
4	Iron Fist	Rare	5 s	Deal 55 damage to enemies within 48 px
5	Blood Siphon	Rare	10 s	Drain the nearest enemy for 45 damage and restore 20 HP
6	Stone Wall	Rare	10 s	Absorb the next 50 damage
7	Remedy Vial	Rare	8 s	Drink a swift remedy restoring 42 HP
8	Nova Burst	Epic	9 s	Deal 110 damage to all enemies within 72 px
9	Mending Wave	Epic	12 s	Restore 85 HP
10	Mirror Guard	Epic	14 s	Gain 58 shield and 1.5 s of invincibility
11	Shadow Blade	Legendary	10 s	Deal 250 damage to enemies within 48 px
12	Phoenix Elixir	Legendary	18 s	Fully restore all HP
13	Diamond Fortress	Legendary	15 s	Absorb the next 100 damage

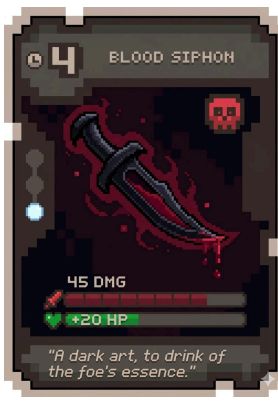
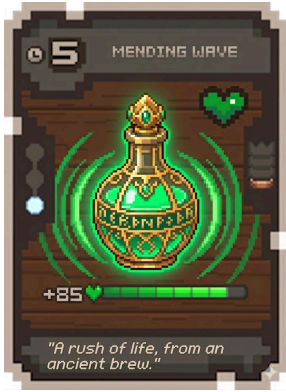
Automatic Cards

#	Name	Rarity	Trigger	Effect
1	Lifetap	Common	On Kill	Restore 20 HP each time you kill an enemy
2	Iron Skin	Common	On Attack	Gain 8 shield each time you hit an enemy
3	Quick Recovery	Common	On Hit Received	Instantly recover 8 HP when struck
4	Wound Echo	Common	On Hit	Deal 10 bonus damage to the struck enemy
5	Rebound	Rare	On Hit Received	Deal 15 damage to enemies within 48 px
6	Berserker Rush	Rare	On Dash	Dashing deals 20 damage to enemies within 32 px
7	Soul Siphon	Rare	On Kill	Killing an enemy restores 18 HP and grants 10 shield
8	Last Stand	Epic	On Hit below 30% HP	Gain 2 s of invincibility
9	Chain Kill	Epic	On Kill	Killing an enemy deals 25 damage to all others within 64 px

2.3.7 Card Design

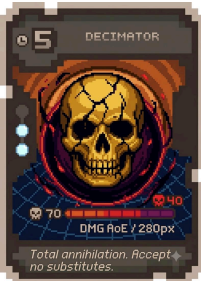
The art of each card conveys its function at a glance. Defense cards use cold, metallic tones; damage cards use warm, aggressive colors; heal cards use warm greens and golds.

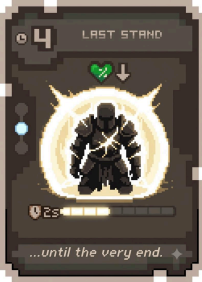
Action Cards — Sample Artwork

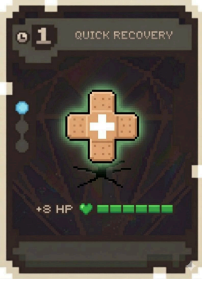
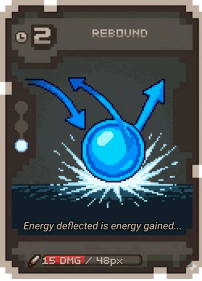
Blood Siphon  4 BLOOD SIPHON 45 DMG +20 HP "A dark art, to drink of the foe's essence."	Diamond Fortress  18 DIAMOND FORTRESS 18 CRYSTALLIZE YOUR BODY. ABSORBING THE NEXT 100 DAMAGE.	Healing Potion  6 HEALING POTION 4 A RESTORATIVE BREW
Iron Fist  4 IRON FIST 5 DMG 55 "A fist forged in ancient iron, glowing with rare energy."	Mending Wave  5 MENDING WAVE +85 "A rush of life, from an ancient brew."	Mirror Guard  5 MIRROR GUARD 58 Reflecting all, deflecting most...
Nova Burst  5 NOVA BURST 110 "A localized collapse, unleashed. The deck demands a grand finale."	Phoenix Elixir  6 PHOENIX ELIXIR FULL HP "A drink of pure creation, from the realm of light."	Quick Strike  2 QUICK STRIKE 25 "A simple, reliable swing to end the fight quickly."

Remedy Vial	Shadow Knife	Stone Wall	Wood Shield
 REMEDY VIAL +42 HP "A more potent, delicate concoction."	 SHADOW KNIFE A BLADEFORGED IN THE DUNGEONS...	 STONE WALL 50 "It stands resolute against the siege."	 WOOD SHIELD 2 A THICK OAK SLAB...

Automatic Cards — Sample Artwork

Aftershock	Berserker Rush	Chain Kill	Decimator
 AFTERSHOCK 40 DMG / 180px The earth always has the last word.	 BERSERKER RUSH 20 DMG / 32px There is no 'before', only 'forward'.	 CHAIN KILL 25 DMG AoE / 64px One break, many fall.	 DECIMATOR 70 DMG AoE / 280px Total annihilation. Accept no substitutes.

Iron Skin	Last Stand	Life Tap	Phantom Step
 IRON SKIN +8 Forced from the ground.	 LAST STAND 25 ...until the very end.	 LIFETAP +20 Death feeds life.	 PHANTOM STEP +22 There is no 'before', only 'forward'. SHIELD

Quick Recovery	Rebound	Soul Siphon	Wound Echo
 QUICK RECOVERY +8 HP	 REBOUND 15 DMG / 48px Energy deflected is energy gained.	 SOUL SIPHON +18 +10 There is no 'before', only 'forward'. SIPHON	 WOUND ECHO +10 DMG

2.4 Combat

2.4.1 Combat Overview

Combat in Dimension Deck is a real-time, continuous experience where enemies spawn and attack immediately upon the player entering a room. The system requires simultaneous attention to spatial awareness and card-based decision-making.

- **Movement and Defense:** Players use WASD or arrow keys for top-down directional control to physically dodge projectiles and melee strikes.
- **The Dodge Mechanic:** The spacebar triggers a dash, providing a brief burst of movement to escape being surrounded.
- **Card Execution:** Players aim with the mouse and execute active cards using number keys 1–5.
- **Telegraphed Danger:** Enemy attacks are telegraphed via wind-up animations or visible paths projected on the floor, giving the player seconds to react.

2.4.2 Enemy Behavioral Archetypes

All enemies are built upon three foundational archetypes that dictate their AI and physical presence.

Swarm Archetype - Small enemies (16 × 16 px) that use a seek-and-close movement pattern. - They pressure the player by crowding the playable space, forcing constant movement to avoid contact damage.

Tank Archetype - Large, high-health sprites that move slowly and are highly resistant to knockback. - They act as obstacles that deal massive contact damage, requiring the player to maintain distance.

Ranged Archetype - Mid-sized enemies that maintain a safe distance from the player. - They fire projectiles on a cooldown, forcing the player to dodge across the room.

2.4.3 Dimension-Specific Bestiary

The Dark Ages

Enemy	Archetype	Stats	Behavior
Dungeon Rat	Swarm	16 HP / 82 spd / 6 dmg	Fast-moving; relies on numbers to corner the player
Skeleton	Tank	90 HP / 38 spd / 18 dmg	High HP; uses slow but heavy telegraphed melee swings
Spirit	Ranged	Medium HP	Fires energy projectiles; phases through obstacles
Two-Headed Giant	Boss	Very high HP	Multi-phase boss with charge and stomp attacks
Skeleton King	Final Boss	900 HP / 32 spd	Phase-based boss; fires skull projectiles at 1.5 shots/s



Skeleton



Spirit



Skeleton King



Two-Headed Giant

Old West

Enemy	Archetype	Stats	Behavior
Desert Rat	Swarm	16 HP / 82 spd / 6 dmg	Aggressive small pests; attack in waves
Bandit	Ranged	30 HP / 52 spd / 8 dmg	Strafes while firing pistol shots at 1.5 shots/s from ~180 px
Minotaur	Tank	High HP	Charges at the player, dealing massive contact damage
Iron Marshal	Final Boss	Very high HP	Phase-based boss; multi-attack pattern combining ranged and melee



Bandit



Minotaur



Brown Rat



Grey Rat



Iron Marshal

2.4.4 Strategic Progression: Card Resistance

To ensure the game remains challenging, the Card Resistance system scales with the player's power. As the player expands their card slots, enemies gain resistance to specific card types, preventing a single strategy from dominating the entire run. See Section 2.3.2 for the full resistance formula.

2.5 Roguelite Progression and Telemetry

Runs are independent, and permanent death resets the player back to the start with their current deck lost. Dimensional credits earned during a run carry over and can be spent on permanent upgrades that expand stats and maximum card capacity across future runs.

Key Performance Indicators tracked:

1. **Time per dimension:** Average minutes spent in Old West vs. Dark Ages per run. Identifies pacing issues and difficulty spikes.
2. **Most discarded card:** Which active card is removed most often at the shop. A frequently discarded card signals it underperforms relative to its cost.
3. **Run length at death:** The room number where most Game Over screens occur. Pinpoints difficulty spikes in the procedural layout.

The game also tracks: win/loss ratio, card usage frequency, credit earnings and spending patterns, and average enemies killed per run. All data is stored per-user in the backend database and is visible to administrators through the Admin Panel.

3. Level Design

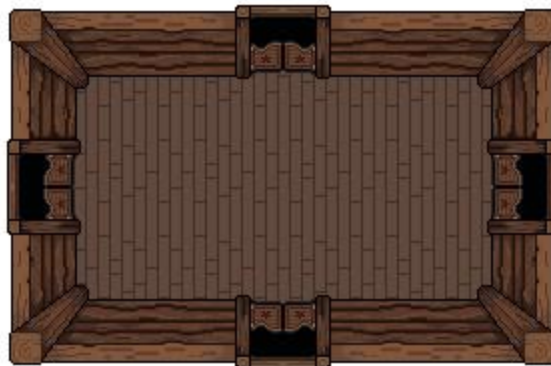
3.1 Themes

3.1.1 Ambience

Dimension Deck uses a 16-bit pixel-art visual style, referencing classics such as *The Legend of Zelda: A Link to the Past* and early SNES RPGs. The game features two distinct dimensions, each with its own look and feel.

The Old West

A dusty frontier dimension with sandy deserts and abandoned outpost towns. The wide-open spaces create a sense of freedom and lawlessness that contrasts sharply with the next world.



Old West Room



Old West Doors



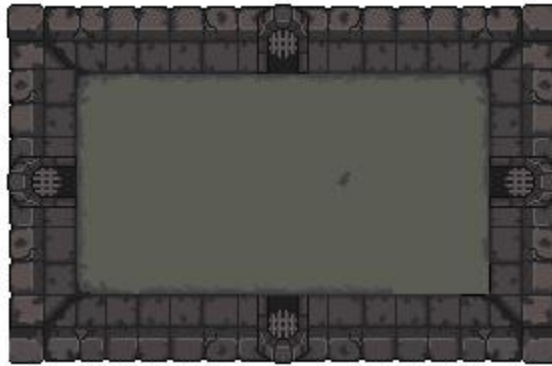
Old West Store



Old West Tiles

The Dark Ages

A fantasy dungeon dimension set inside a castle and caverns. Inspired by *Dungeons & Dragons*, the enclosed stone walls and dim lighting create a feeling of constant threat and confinement.



Dungeon Room



Dungeon Doors



Dungeon Store



Dungeon Tiles

3.1.2 Objects

3.1.2.1 Ambient

- **Player Character:** 32×32 px sprite. Larger size makes the hero stand out against the background and enemies.
- **Theme Elements:** Decorative items (desert plants, stone pillars, torches) ground the player in each dimension.

3.1.2.2 Interactive

Object	Interaction	Effect
Card Reward Chest	Press E after room clear	Draws 1 card; disappears if skipped
Dimension Store	Automatic on enter	Rotating stock of 5 cards for sale; also allows selling and slot upgrades
Glitch Pillar	Walk into it	Instantly grants 1 random card (no selection screen)
Portal	Walk into it	Transitions to the next dimension after the boss is defeated



Glitch Pillar



Portal



Coin

3.1.3 Challenges

- **Real-Time Combat:** Enemies move and attack the moment the player enters the room. The player must constantly dodge or block incoming projectiles and melee hits while playing cards to deal damage.
- **Attack Visualization:** Most enemies telegraph attacks through animations or visible floor markers, giving the player seconds to react.
- **Card Resistance:** As the deck grows, enemies develop resistances, forcing the player to change strategy throughout the run.
- **Boss Battles:** Each dimension ends with a boss fight. The boss changes behavior between phases, testing both deck quality and mechanical skill.
- **Procedural Rooms:** Every room is randomized — the player never knows if the next room contains dangerous enemies, traps, loot, or the level boss.
- **Permanent Death:** Losing all health ends the run and the current deck is lost, requiring a fresh start.

3.2 Game Flow

1. **Start of Run:** A run begins in one of the two dimensions. The player enters the first room with a starter deck (Quick Strike, Heal Pulse, Wood Shield), full directional control, and 100 HP.
2. **Clearing Rooms:** To move forward, the player must eliminate all enemies in a combat room or complete the room's objective. Cleared rooms drop dimensional credits or cards.
3. **Dimension Map:** Between rooms the player sees the branching map and selects their next path.

4. **Shop:** Shop rooms appear periodically, allowing the player to buy cards, sell cards, or upgrade card slots.
 5. **Final Boss Encounter:** At the end of the dimension's node path, the player fights the Final Boss. Defeating it opens a portal to the next dimension.
 6. **Dimension Transition:** The portal transports the player into the new dimension with different enemies, visuals, and card drop pools. Credits and deck contents carry over.
 7. **Victory / Defeat:** Defeating both dimension bosses completes the run and triggers the Victory screen. Losing all HP at any point triggers the Defeat screen.
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4. Development

The codebase is organized around a set of abstract base classes that define shared behavior, keeping individual systems extensible and the overall architecture clean.

- **Entity:** Root class for anything that exists in the game world. Holds position, velocity, health, and a basic update/render cycle.
- **Card:** Defines the structure every card must follow: name, rarity, cooldown, effect function, and card type flag (active or automatic).
- **Enemy:** Extends Entity. Adds abstract methods for movement behavior, attack patterns, a Card Resistance value, and a drop table reference.
- **Room:** Container for a playable space — fixed 480×352 px across a 15×11 tile grid — with spawn points, exit conditions, and lists of enemies and objects.
- **Dimension:** Groups a set of room types, an enemy pool, a visual theme, and a boss reference into a single traversable dimension.

4.1 Derived Classes

Player (extends Entity) - Rendered at 32×32 px. - Manages the active card slot system (3 slots, expandable to 5), the automatic card pool (4–8 slots), and dimensional credit tracking. - Handles wall collision, mouse-based aim direction, and fires automatic card triggers on combat events (dealing damage, killing an enemy, taking damage, or dashing).

ActiveMeleeCard, ActiveHealCard, ActiveDefenseCard, ActiveDrainCard (extend ActiveCard) - *ActiveMeleeCard*: Cone-shaped area damage to all enemies within a configurable range and spread angle. - *ActiveHealCard*: Restores a fixed HP amount; `null` heal amount triggers a full restore. - *ActiveDefenseCard*: Grants flat shield and optional invincibility frames. - *ActiveDrainCard*: Finds the nearest living enemy, damages it, and returns a portion as healing.

AutomaticCard (extends Card) - Sits in the automatic pool and fires its effect when a trigger condition is met, without player input. - Supported triggers: `on_attack`, `on_kill`, `on_hit_received`, `on_dash`. - All automatic cards are

defined as object instances in `CardCatalog.js` rather than individual subclasses.

SwarmEnemy, TankEnemy, RangedEnemy (extend Enemy) - *SwarmEnemy*: Seek-and-close movement; deals contact damage. - *TankEnemy*: Slow movement; very high HP and contact damage. - *RangedEnemy*: Maintains distance; fires projectiles at a set rate and preferred distance.

Dimension-specific enemies (extend archetypes) - Dark Ages: *DungeonRat* (Swarm), *FallenKnight* (Tank), *Spirit* (Ranged), *SkeletonKing* (Boss) - Old West: *DesertRat* (Swarm), *Minotaur* (Tank), *Bandit* (Ranged), *TheIronMarshal* (Boss)

BossEnemy (extends Enemy) - Adds a phase list and a phase transition trigger. - Boss sprites are larger than standard enemies (80×80 px for *SkeletonKing*). - Each dimension has its own Final Boss.

CombatRoom, ShopRoom, ChestRoom, BossRoom, GlitchRoom, ShrineRoom (extend Room) - Each overrides spawn logic and exit conditions within the fixed 480×352 px, 15×11 tile grid. - *CombatRoom*: Spawns enemies from the dimension pool; unlocks exits on full clear. - *ShopRoom*: Spawns the merchant NPC; pre-clears itself; provides the interactive store overlay. - *ChestRoom*: Spawns one chest; exits after interaction. - *BossRoom*: Spawns the dimension Final Boss; fires the dimension-advance callback on defeat. - *GlitchRoom*: Contains a Glitch Pillar that grants a random card on contact. - *ShrineRoom*: Offers a single card upgrade trade.

5. Graphics

5.1 Style Attributes

The game uses a **16-bit pixel-art aesthetic**, drawing from Super Nintendo titles such as *The Legend of Zelda: A Link to the Past*, *The Escapists*, and the *Mario & Luigi* saga. Each dimension has a unique visual identity that provides clear spatial feedback.

- **Resolution:** 16×16 px per tile.
- **Room Size:** 480×352 px (15×11 tile grid).
- **Style:** Pixel art in 2D top-down perspective.

HUD Layout - Cards (bottom center): The active card hand displayed in the lower central area. Selected cards are highlighted. - **Player HP (bottom right):** Current health and shield values. - **Deck Menu (bottom left):** Opens the full deck for slot management and card review.

Visual Feedback - When the player takes damage, the screen flashes red briefly. - Enemies briefly turn red when hit. - Card acquisition displays the card image prominently to draw the player's attention.

5.2 Sprites

Player Characters

Knight	Cowboy
	

Both player characters are rendered at 32×32 px and include: - Idle animations in 4 directions (left, right, up, down) - Walking animations in 4 directions - Take-damage animation - Death animation

Enemy Sprites (Dark Ages)

Skeleton	Spirit	Skeleton King	Two-Headed Giant
			

Enemy Sprites (Old West)

Bandit	Minotaur	Brown Rat	Grey Rat	Iron Marshal
				

Tiles and Rooms

Dark Ages Tiles	Old West Tiles
	

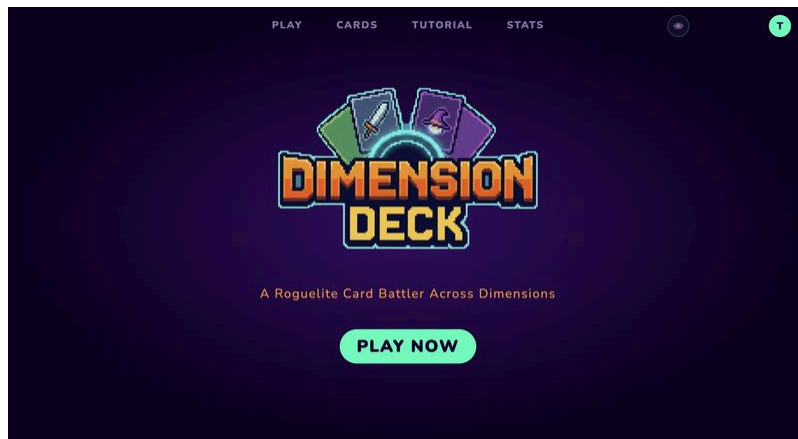
6. Web Platform

Dimension Deck includes a full web platform that serves as the player's hub for account management, statistics, and game access. The platform is built with Vanilla JavaScript and a Node.js/Express backend connected to a MySQL database.

6.1 Pages

Main Menu

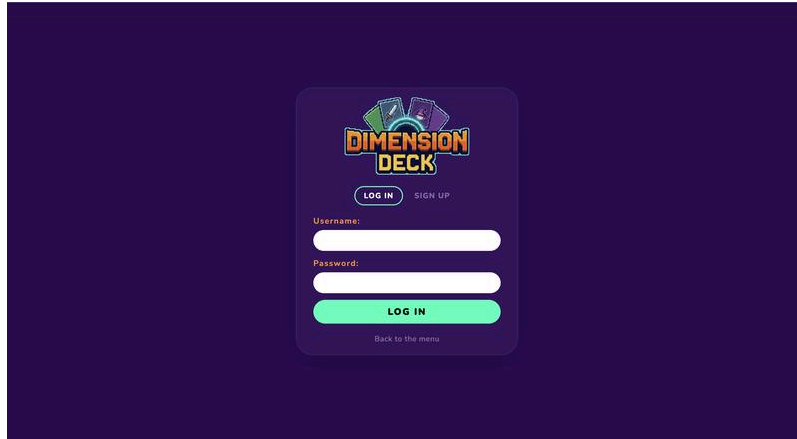
The landing page where players can access the game, view their profile, and navigate the site.



Main Menu

Login / Sign Up

Players create accounts or log in to track their runs and scores.



Login & Signup

Cards Page

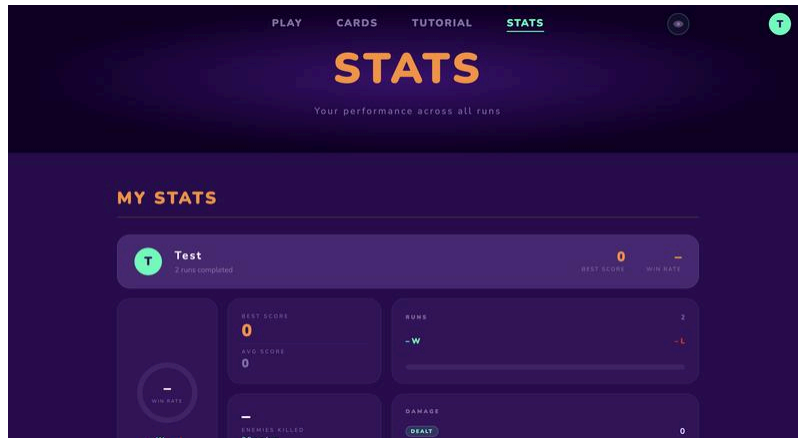
A browseable catalog of all available cards in the game, including stats, rarity, and type.



Cards

Stats Page

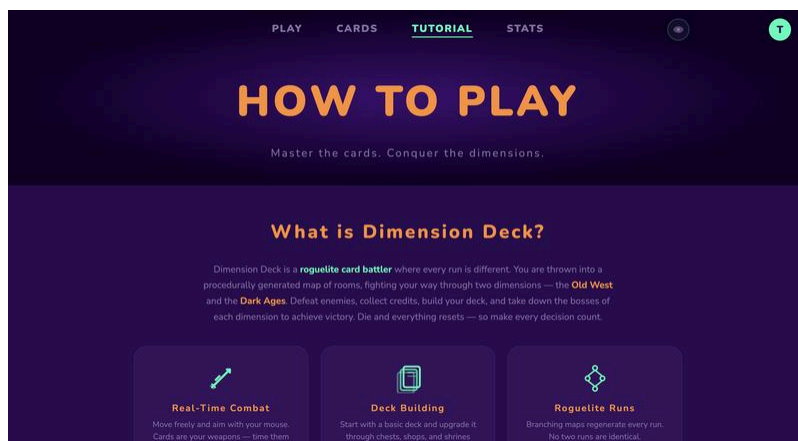
Personal statistics dashboard for logged-in players: win rate, best score, damage dealt, rooms cleared, card history, and leaderboard.



Stats

Tutorial Page

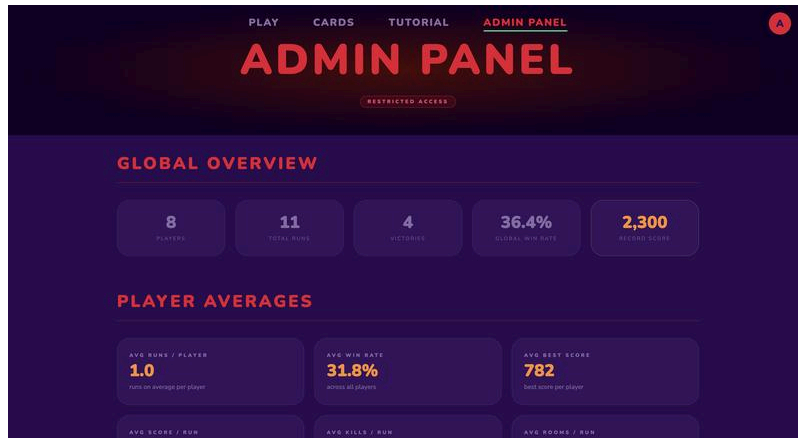
An interactive guide explaining game mechanics, controls, and card types.



Tutorial

Admin Panel

Restricted to admin accounts. Displays aggregate statistics across all players: total runs, global win rate, record score, per-player leaderboard, and performance charts.



Admin Panel

6.2 Logo



Dimension Deck Logo

7. Sounds / Music

7.1 Style Attributes

The soundtrack is a core component of the game experience, built around a 16-bit aesthetic using synthesized sound and short looping compositions that reinforce the repetitive nature of a roguelite. The primary goal of the soundtrack is to generate constant tension, communicating that the player is in a dangerous environment where enemies can appear at any moment.

Each dimension has its own distinct soundtrack:

- **Old West:** Inspired by classic western desert styles. Slow tempos; conveys isolation, vastness, and latent danger.
 - Reference: <https://www.youtube.com/watch?v=KhChHmk1o8c>
- **Dark Ages:** Inspired by dark fantasy dungeon settings and classic games such as *The Legend of Zelda*. Conveys confinement and constant threat.
 - Reference: <https://www.youtube.com/watch?v=Nxx3Ti83TYk>

7.2 Sounds Needed

All sound effects follow a 16-bit retro aesthetic.

UI Sounds - Card reveal - Card played - Menu selection —
<https://www.freesound.org/people/ZeltBolt/sounds/833055/>

Interaction Sounds - Chest opened - Glitch Pillar activated - Shop purchase / sell - Card upgrade merge

Combat Sounds - Player takes damage - Enemy takes damage / dies - Dash / dodge - Projectile fired / impact

Ambient Sounds - Desert wind —
<https://www.freesound.org/people/felix.blume/sounds/156414/> - Whispering distant voices (Dark Ages)

7.3 Music Needed

Context	Track Reference
Dungeon exploration	https://www.newgrounds.com/audio/listen/625484
Combat	https://www.newgrounds.com/audio/listen/76637
Final boss battle	https://www.newgrounds.com/audio/listen/436762
Main menu	https://www.playonloop.com/2020-music-loops/misty-dungeon/
Defeat screen	https://www.newgrounds.com/audio/listen/368887
Victory screen	https://www.newgrounds.com/audio/listen/1323003
Credits screen	https://www.newgrounds.com/audio/listen/11684

8. Schedule

Week	Milestone	Tasks
1	Core System	Entity and player base classes; basic movement and collision; canvas rendering pipeline
2	Gameplay & Card System	Active and automatic card classes; cooldown system; CardCatalog; starter deck
3	Procedural Map Design	Room generator; branching dimension map; room type weighting; Glitch Pillar and Chest logic
4	Enemies & Combat	Swarm, Tank, and Ranged archetypes; dimension-specific enemies; boss phase system; Card Resistance scaling
5	UI / UX	HUD (health, shield, card slots, minimap); Deck Management screen; Pause menu; Shop UI
6	Web Platform	Node.js/Express backend; MySQL schema; authentication (JWT); player stats API; leaderboard
7	Integration & Polish	Frontend–backend integration; admin panel; telemetry endpoints; sprite polish; audio integration
8	Testing & Delivery	Full playtest; balance pass; bug fixes; documentation (GDD,

Week	Milestone	Tasks
		SRS, UML); final submission